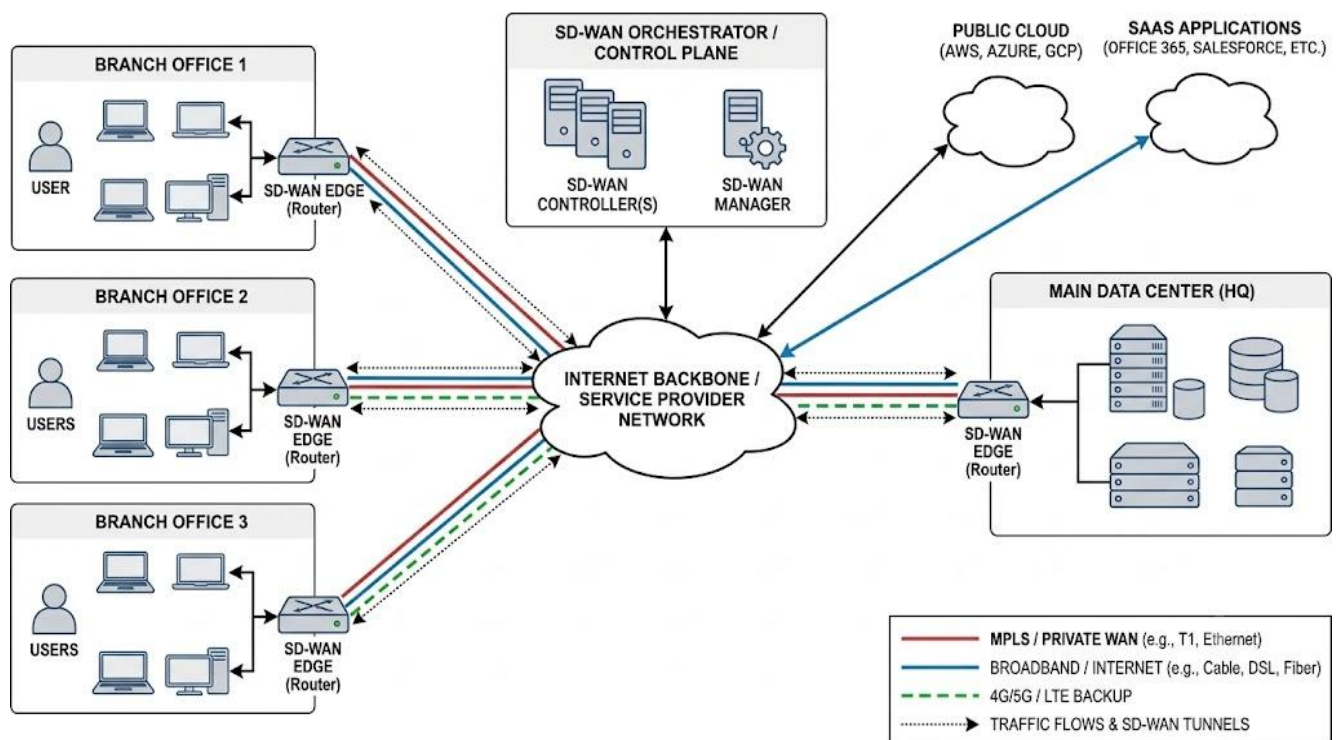


Task-1

Feature	Traditional WAN	SD-WAN
1. Data Routing	Traffic usually follows predefined routes through dedicated WAN links.	Traffic can be dynamically routed over MPLS, broadband, or 4G/5G based on network conditions.
2. Management	Managed mainly through individual routers and manual configuration.	Centrally managed using software/controllers and a single management interface.
3. Security	Security often relies on separate firewalls, VPNs, and security appliances.	Security policies can be centrally configured and integrated with encryption, firewalls, and other security features.
4. Network Flexibility	Adding or changing WAN links can require significant manual configuration.	New links and sites can generally be configured more easily through centralized software management.
5. Traffic Control	Limited ability to automatically prioritize applications.	Can identify applications and prioritize important traffic such as video conferencing or business applications.
6. Cost	Often depends heavily on expensive dedicated connections such as MPLS.	Can combine lower-cost Internet connections with other WAN links, potentially reducing connectivity costs.

Task-2



Task-3

Scenario: Instagram Using SD-WAN

Imagine Instagram operates data centers and offices across multiple cities in India, such as **Mumbai, Delhi, Bengaluru, and Ahmedabad**. SD-WAN could automatically select the best available network path based on speed, congestion, and reliability. For example, if one Internet connection in Mumbai becomes slow, SD-WAN can route traffic through another available connection. It can also use encryption and centralized security policies to protect data. This can provide users with **faster loading, fewer interruptions, better reliability, and more secure access** to Instagram services from different cities.

Conclusion: SD-WAN can improve the user experience by providing **dynamic routing, reliable connectivity, and centralized security** across multiple locations.

Task-4

Three Key Benefits of SD-WAN

Benefit	Explanation	Real-world Example
1. Better Performance	SD-WAN can automatically select the best available network path based on network conditions.	Employees in the Ahmedabad office using video meetings with the Mumbai office can be routed through the better connection, reducing delays and interruptions.
2. Higher Reliability	SD-WAN can use multiple connections and automatically switch to another link if one fails.	If the main Internet connection at the Bangalore office goes down, traffic can automatically move to a backup connection, allowing employees to continue working.
3. Improved Security & Centralized Management	Security policies can be centrally managed and applied consistently across all offices.	The company can apply the same access and security rules in Ahmedabad, Mumbai, and Bangalore , helping protect customer data while employees access company applications.

Conclusion: SD-WAN helps a multi-city company achieve **better network performance, reliable connectivity, and centralized security management.**